

SCOPE · B.TECH PROJECT

Adaptive Physics-Informed Battery Digital Twin for SOH and RUL Prediction

Energy AI · Battery Management Systems · Physics-Informed Neural
Networks

SOH · RUL

TEAM

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DOMAIN & MOTIVATION

Project Overview & Use Cases

WHERE IT APPLIES

DOMAIN

Energy AI · Battery Management Systems

TYPE OF DNN

Physics-Informed Neural Network (PINN) +
Feedforward DNN

THE CHALLENGE

Li-ion batteries degrade with cycling,
temperature and age. Pure deep-learning
models need large labeled datasets and can
be physically inconsistent; pure physics
models are too slow for real-time use.

OUR APPROACH

An Adaptive PINN Battery Digital Twin fuses
PINNs, a Transformer Encoder and
Incremental Adaptive Learning to
continuously estimate SOH and RUL —
adapting to new data without full retraining.

FOUNDATION REFERENCE

Base Paper

Physics-Informed Neural Networks for Battery Modeling and State Estimation: A Methodological Review

Renewable and Sustainable Energy Reviews, 2026

WHY THIS PAPER WAS SELECTED

Comprehensive review of Physics-Informed Neural Networks (PINNs) for battery applications

Covers SOH, RUL, SOC, Digital Twins, thermal diagnosis and internal state reconstruction

Explains multiple methods for embedding battery physics into neural networks

Discusses training stability, domain generalization, uncertainty quantification and interpretability

Lays out future research directions for intelligent Battery Management Systems

POSITIONING OUR WORK

Literature Review & Research Gap

RESEARCH GAP

Existing PINN models are trained offline, adapt poorly to changing battery behaviour, and remain difficult to deploy in real-time BMS. Lightweight, continual-learning methods are still an open problem.

REAL-WORLD IMPACT
Applications

1

Electric Vehicles

Predict battery replacement before failure occurs.

2

Solar Storage

Monitor home batteries and predict maintenance needs.

3

Smart Grids

Avoid sudden, costly battery failures at scale.

4

Drones

Estimate safe remaining flying time in-flight.

5

Laptops & Smartphones

More accurate battery degradation estimates.

NASA cells and a leakage-safe evaluation protocol

The official NASA PCoE repository provides the first committed dataset for this proof of concept.

EOL RULE	HELD-OUT CELL	CYCLES PARSED	ORDER	METRIC UNIT
70% SOH	B0018	636 total	Time-aware	RUL in cycles

From measured cycle to adaptive digital-twin state

The first-stage physics model uses relationships supported by the NASA measurements.

NASA records

Cycle-level voltage, current, temperature, time, capacity

Masked encoder

Measured points + delta_t, no blind interpolation

Temporal model

History across cycles with Transformer context

SOH / RUL heads

Bounded SOH 0–100 and non-negative RUL

Physics losses

Soft degradation, RUL order, capacity-curve residual

Drift trigger, recent + representative history

Training objective

The SEI-growth PDE formulation remains a Stage 2 extension until every variable and parameter is available and verified.

The active training objective and physical diagnostics

The first-stage POC uses capacity/degradation consistency that the NASA records can support; it does not claim an SEI-PDE residual.

selected: $\lambda_{\text{phys}} = 0.1 \cdot \lambda_{\text{cons}} = 1 \cdot \text{RUL scale} = 250 \text{ cycles}$

DATA LOSS

$$L_{\text{data}} = \text{mean}|\hat{\text{soh}} - \text{soh}| + \text{mean}|\hat{\text{rul}} - \text{rul}|/250$$

Joint SOH and RUL regression with valid-cycle masks.

CURVE CONSISTENCY

$$L_{\text{cons}} = \text{mean}|\Delta\hat{\text{soh}} - \Delta\text{soh}| / 100$$

Capacity-curve shape residual between predicted and measured SOH changes.

SOFT PHYSICS LOSS

$$L_{\text{phys}} = \text{mean ReLU}(\Delta\hat{\text{soh}}) + \text{mean ReLU}(\Delta\hat{\text{rul}})$$

Penalizes predicted SOH or RUL increases across ordered cycles; soft because real cells can show local regeneration.

BOUNDED HEADS

$$\hat{\text{soh}} = 100 \cdot \sigma(z_1), \hat{\text{rul}} = 250 \cdot \sigma(z_2)$$

Output contract: SOH $\in [0,100]$ and RUL $\in [0,250]$. The nonnegative-consistency term is an explicit zero-valued future hook.

Diagnostics reported: SOH bounds violation · negative RUL rate · degradation violation rate · SOH/RUL consistency violation · capacity-curve residual

Transcribed from the methodological review, pp. 4–5. These are representative framework equations, not one new end-to-end model.

(1) NEURAL APPROXIMATION

A network represents a battery state variable such as concentration, voltage, temperature, or SOC.

(3) RESIDUAL

Automatic differentiation evaluates the physical-law residual at a point.

(5) COMPOSITE PINN LOSS

Weights balance data fit, governing physics, boundary/initial conditions, and optional regularization.

(2) GOVERNING OPERATOR

$\mathcal{N}$ is the physical operator derived from electrochemical, ECM, aging, or thermal equations; λ denotes physical parameters.

(4) PHYSICS LOSS

Residuals are averaged over N_p physics collocation points.

Implementation: We specialize data loss, physical monotonicity, and capacity curve residual losses for battery health tracking.

The selected configuration came from the complete grid

Every requested combination ran across seeds 7, 17, and 27 before the final evidence pass.



Selected configuration

The final evidence runner then used the selected configuration on the full default cycle-window representation.

GRID STATUS

54 / 54 complete

MODEL COUNT

5 comparisons

EVIDENCE MODE

Held-out cell

OUTPUT

JSON + CSV + PNG

The standalone PINN led frozen SOH accuracy

All model rows use the same B0018 held-out evaluation protocol and four training epochs.

Adaptive twin

Best frozen SOH row

PINN: 7.76 ± 1.42 SOH MAE

Measured model comparison under the held-out-cell protocol

Mean $\pm$ standard deviation over seeds 7, 17, and 27; B0005/B0006/B0007 train and B0018 held out.

Readout

The strongest evidence is the low-cost online update. The main limitation is the small four-cell dataset.

What the run supports

80.63% lower post-shift SOH MAE than frozen.

8.75% of full-retraining work.

What the run does not prove

- Universal chemistry or temperature generalization
- Perfect monotonic degradation behavior
- SEI-PDE validity without additional variables
- A direct numerical ranking against the four papers

Next experiment

Adaptation targets, ablations, and robustness results

All required experiment families were executed; the acceptance table is the direct proof-of-concept decision record.

Ablation post-shift SOH MAE

Physical checks: SOH bounds 0.00% · negative RUL 0.00% · capacity-curve residual 0.00573 ± 0.00009 · *negative forgetting means prior-domain MAE improved

CONCLUSION

Project Summary

Proposed an Adaptive Physics-Informed Battery Digital Twin for accurate SOH and RUL estimation.

Integrated Transformer Encoder, PINN and Feedforward DNN to combine data-driven learning with battery physics.

Introduced incremental adaptive learning to keep the Digital Twin current without full model retraining.

Overcomes key limits of pure deep-learning and pure physics-based models with real-time, physically consistent predictions.

Potential impact: Electric Vehicles, Battery Management Systems, Renewable Energy Storage, Smart Grids, Drones and Consumer Electronics.